Problem 1) $\begin{cases} 2\ddot{x} + \omega^2(5x-3y) = 0 & \text{(I)} \\ 2\ddot{y} + \omega^2(5y-3x) = 0 & \text{(II)} \end{cases}$ Realizing the symmetry of the problem, we can apply a trick to solve it!

$I + II = 2(\ddot{x}+\ddot{y}) + 2\omega^2(x+y) = 0 \quad ; \quad z = x+y$

$\Rightarrow \ddot{z} + \omega^2 z = 0 \Rightarrow z(t) = x(t)+y(t) = 2A_1\cos(\omega t-\delta_1)$

$I - II = 2(\ddot{x}-\ddot{y}) + 8\omega^2(x-y) = 0 \quad ; \quad \eta = x-y$

$\Rightarrow \ddot{\eta} + 4\omega^2\eta = 0 \Rightarrow \eta(t) = x(t)-y(t) = 2A_2\cos(2\omega t-\delta_2)$

$\Rightarrow z(t)+\eta(t) = 2x(t) = 2A_1\cos(\omega t-\delta_1) + 2A_2\cos(2\omega t-\delta_2)$

$\Rightarrow \boxed{x(t) = A_1\cos(\omega t-\delta_1) + A_2\cos(2\omega t-\delta_2)}$

$\Rightarrow z(t)-\eta(t) = 2y(t) = 2A_1\cos(\omega t-\delta_1) - 2A_2\cos(2\omega t-\delta_2)$

$\Rightarrow \boxed{y(t) = A_1\cos(\omega t-\delta_1) - A_2\cos(\omega t-\delta_2)}$

We were lucky that this trick worked. In general, we are not so lucky, but we can always try our best guess solution to see if it works. For example, because the nature of the equations is oscillatory, we can try a solution of the form $x = Ae^{i\alpha t}; y = Be^{i\alpha t}$. We can absorb i in α, but that does not make any difference other than making our solutions ugly in the end. Then,

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix} e^{i\alpha t}$. Substituting this eqn into the coupled eqns of the problem yields,

$\begin{cases} 2A(-\alpha^2) + 5A\omega^2 - 3B\omega^2 = 0 \\ 2B(-\alpha^2) + 5B\omega^2 - 3A\omega^2 = 0 \end{cases}$ we can write this in matrix form as,

$\begin{pmatrix} -2\alpha^2+5\omega^2 & -3\omega^2 \\ -3\omega^2 & -2\alpha^2+5\omega^2 \end{pmatrix}\begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{(III)}$

This indicates that the determinant of the 2x2 matrix of coefficients must be zero for eqn III to hold.

$\Rightarrow 4\alpha^4 - 20\alpha^2\omega^2 + 16\omega^4 = 0 \Rightarrow \begin{cases} \alpha = \pm\omega \Rightarrow A = B \\ \alpha = \pm 2\omega \Rightarrow A = -B \end{cases}$

Adding up all four eqns for various values of α yields,

$\begin{pmatrix} x \\ y \end{pmatrix} = A_1\begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{i\omega t} + A_2\begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{-i\omega t} + A_3\begin{pmatrix} 1 \\ -1 \end{pmatrix}e^{2i\omega t} + A_4\begin{pmatrix} 1 \\ -1 \end{pmatrix}e^{-2i\omega t}$

The A_i are determined by the initial conditions. If x and y are positions of real particles, then A_1 and A_2, and A_3 and A_4 must be complex conjugates so that the imaginary components of the solution cancel out, effectively yielding:

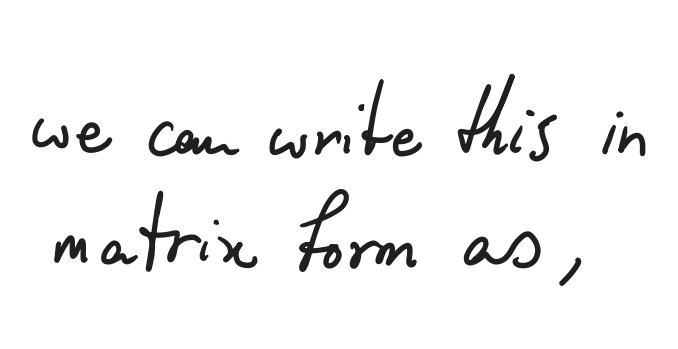
$\begin{pmatrix} x \\ y \end{pmatrix} = B_1\begin{pmatrix} 1 \\ 1 \end{pmatrix}\cos(\omega t-\delta_1) + B_2\begin{pmatrix} 1 \\ -1 \end{pmatrix}\cos(2\omega t-\delta_2)$

The vectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are the eigenvectors of the system's oscillatory behavior with the corresponding eigenvalues ω and 2ω. The variables $z = x+y$ and $\eta = x-y$ are called the normal-mode oscillations of the system.

Problem 2) For the moment, let's assume the middle spring has a different k_{12}.

The force on m_1 is: $F_1 = -kx_1 - k_{12}(x_1-x_2)$

The force on m_2 is: $F_2 = -kx_2 - k_{12}(x_2-x_1)$



Thus, the equations of motion are:

$\begin{cases} m\ddot{x}_1 + (k+k_{12})x_1 - k_{12}x_2 = 0 \\ m\ddot{x}_2 + (k+k_{12})x_2 - k_{12}x_1 = 0 \end{cases} \quad \text{(I)}$

Because we expect the motion to be oscillatory, we attempt a solution of the form,

$\begin{cases} x_1 = B_1e^{i\omega t} \\ x_2 = B_2e^{i\omega t} \end{cases} \quad \text{(II)}$

Because these solutions are complex, we will only keep the real parts in the end. Substituting yields,

$\begin{cases} -m\omega^2B_1e^{i\omega t} + (k+k_{12})B_1e^{i\omega t} - k_{12}B_2e^{i\omega t} = 0 \\ -m\omega^2B_2e^{i\omega t} + (k+k_{12})B_2e^{i\omega t} - k_{12}B_1e^{i\omega t} = 0 \end{cases} \quad \text{(III)}$

$\Rightarrow \begin{cases} (k+k_{12}-m\omega^2)B_1 - k_{12}B_2 = 0 \\ -k_{12}B_1 + (k+k_{12}-m\omega^2)B_2 = 0 \end{cases} \quad \text{(IV)}$

$\Rightarrow \begin{vmatrix} k+k_{12}-m\omega^2 & -k_{12} \\ -k_{12} & k+k_{12}-m\omega^2 \end{vmatrix} = 0 \quad \text{(V)}$

$\Rightarrow (k+k_{12}-m\omega^2)^2 - k_{12}^2 = 0 \Rightarrow k+k_{12}-m\omega^2 = \pm k_{12}$

$\Rightarrow \omega = \sqrt{\frac{k+k_{12}\pm k_{12}}{m}} \Rightarrow \begin{cases} \omega_1 = \sqrt{\frac{k+2k_{12}}{m}} \\ \omega_2 = \sqrt{\frac{k}{m}} \end{cases} \quad \text{(VI)}$

That is, the system has two characteristic frequencies (or eigen-frequencies). Thus, the general solution to the problem is,

$x_1(t) = B_{11}^+e^{i\omega_1t} + B_{11}^-e^{-i\omega_1t} + B_{12}^+e^{i\omega_2t} + B_{12}^-e^{-i\omega_2t} \quad \text{(VII)}$

$x_2(t) = B_{21}^+e^{i\omega_1t} + B_{21}^-e^{-i\omega_1t} + B_{22}^+e^{i\omega_2t} + B_{22}^-e^{-i\omega_2t}$

where the amplitudes relationships can be determined by substituting the solutions into the eqns (IV) as,

$\begin{cases} B_{11} = -B_{21} & \text{for } \omega = \omega_1 \\ B_{12} = B_{22} & \text{for } \omega = \omega_2 \end{cases}$

Thus, the only necessary subscripts for Bs are the ones for the eigenvalues,

$\begin{cases} x_1(t) = B_1^+e^{i\omega_1t} + B_1^-e^{-i\omega_1t} + B_2^+e^{i\omega_2t} + B_2^-e^{-i\omega_2t} \\ x_2(t) = -B_1^+e^{i\omega_1t} - B_1^-e^{-i\omega_1t} + B_2^+e^{i\omega_2t} + B_2^-e^{-i\omega_2t} \end{cases}$

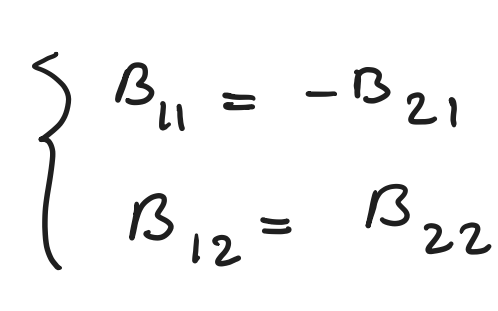
Thus, we have four initial conditions for two second-order ODE, which makes sense. If we define the following new set of coordinates,

$\begin{cases} \eta_1(t) = x_1 - x_2 \\ \eta_2(t) = x_1 + x_2 \end{cases}$, we can rewrite (I) as,

$\begin{cases} m(\ddot{\eta}_1+\ddot{\eta}_2) + (k+2k_{12})\eta_1 + k\eta_2 = 0 \\ m(\ddot{\eta}_1-\ddot{\eta}_2) + (k+2k_{12})\eta_1 - k\eta_2 = 0 \end{cases}$ Adding and subtracting these two equations yields,

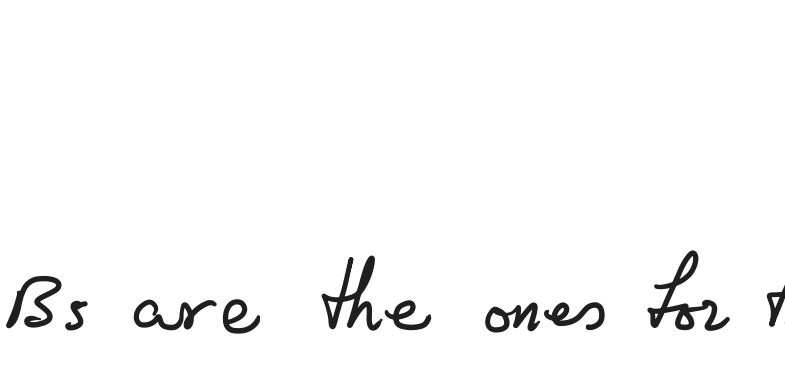
$\begin{cases} m\ddot{\eta}_1 + (k+2k_{12})\eta_1 = 0 \\ m\ddot{\eta}_2 + k\eta_2 = 0 \end{cases}$ We now have two separate eqns!

$\Rightarrow \begin{cases} \eta_1(t) = C_1^+e^{i\omega_1t} + C_1^-e^{-i\omega t} \\ \eta_2(t) = C_2^+e^{i\omega_2t} + C_2^-e^{-i\omega_2t} \end{cases}$ Thus, η_1 and η_2 are the normal coordinates.



$\omega = \omega_1$

asymmetrical mode
(out of phase)



symmetrical mode
(in phase)

We can always set the initial conditions such that only one normal mode is excited as in the above figures.

Problem 3) $U = \frac{1}{2}kx_1^2 + \frac{1}{2}k_{12}(x_2-x_1)^2 + \frac{1}{2}kx_2^2$

$= \frac{1}{2}(k+k_{12})x_1^2 + \frac{1}{2}(k+k_{12})x_2^2 - k_{12}x_1x_2$

the term proportional to x_1x_2 is the factor that describes the coupling.

$\Rightarrow \begin{cases} A_{11} = \left.\frac{\partial^2U}{\partial x_1^2}\right|_0 = k+k_{12} \\ A_{12} = \left.\frac{\partial^2U}{\partial x_1\partial x_2}\right|_0 = -k_{12} = A_{21} \\ A_{22} = \left.\frac{\partial^2U}{\partial x_2^2}\right|_0 = k+k_{12} \end{cases}$

$T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 = \frac{1}{2}\sum_{j,k} m_{jk}\dot{x}_j\dot{x}_k \Rightarrow \begin{cases} m_{11} = m_{22} = m \\ m_{12} = m_{21} = 0 \end{cases}$

The secular determinant becomes,

$\begin{vmatrix} k+k_{12}-m\omega^2 & -k_{12} \\ -k_{12} & k+k_{12}-m\omega^2 \end{vmatrix} = 0$

This yields the same equation as we got using the Newtonian approach. Thus, $\omega = \sqrt{\frac{k+k_{12}\pm k_{12}}{m}} \Rightarrow \omega_1 = \sqrt{\frac{k+2k_{12}}{m}}; \omega = \sqrt{\frac{k}{m}}$

Problem 4) Let x_1 and x_2 represent the displacements of the two masses from the equilibrium point. Assume the driving force acts on the mass ①, then $F = ma$ eqns are:

$\text{(I)} \begin{cases} m\ddot{x}_1 + 2k(x_1-x_2) = F_d\cos\omega_dt \\ m\ddot{x}_2 + 2k(x_2-x_1) = 0 \end{cases}$ assuming x_1 & x_2 are measured counterclockwise.

To solve the above eqns, we can treat $F_d\cos\omega_dt$ as the real part of the $F_de^{i\omega_dt}$ and try solutions of the form,

$x_1(t) = A_1e^{i\omega_dt}$; $x_2(t) = A_2e^{i\omega_dt}$ and solve for A_1 & A_2.

or taking the real part, $x_1(t) = A_1\cos\omega_dt$; $x_2(t) = A_2\cos\omega_dt$

Combined with (I) this yields, $\begin{cases} -\omega_d^2A_1 + 2\omega^2(A_1-A_2) = F \\ -\omega_d^2A_2 + 2\omega^2(A_2-A_1) = 0 \end{cases}$ $\omega \equiv \sqrt{\frac{k}{m}}$ $F = F_d/m$

Solving the above for A_1 and A_2, we get

$x_1(t) = \frac{-F(2\omega^2-\omega_d^2)}{\omega_d^2(4\omega^2-\omega_d^2)}\cos\omega_dt$; $x_2(t) = \frac{-2F\omega^2}{\omega_d^2(4\omega^2-\omega_d^2)}\cos\omega_dt$

Problem 5) Since the relaxed length is zero, the spring force can be written as $-k\vec{r}$ (where $\vec{r}$ is measured relative to the fixed end), because the vector $-\vec{r}$ contains the correct magnitude and direction. Moving one spring from $\vec{r}$ to $\vec{r}_2$ would create a net force difference of $\Delta\vec{F} = -k(\vec{r}_2-\vec{r}_1) \equiv -k\Delta\vec{r}$. The same statement is true about the other spring. Thus, the net force felt by the mass after $\Delta\vec{r}$ displacement is $\Sigma\Delta F = -2k\Delta r = -(2k)\Delta r$, which is equivalent to the force of a single spring with constant $2k$. Thus, the motion is harmonic oscillation on a straight line (determined by the direction from the equilibrium point to the initial position) with frequency $\sqrt{\frac{2k}{m}}$, independent of the direction of the initial kick.

Problem 6) The same argument as in problem ⑤ can be applied to write $\vec{F} = -(k_1+k_2+\cdots+k_n)\Delta\vec{r}$, which means the motion is a harmonic oscillator with frequency $\sqrt{(k_1+\cdots+k_n)/m}$ independently of the direction of the initial kick.